

Course Overview

As Artificial Intelligence (AI) becomes integrated into high-risk domains like healthcare, finance, and criminal justice, it is critical that those responsible for building these systems think outside the black box and develop systems that are not only accurate, but also transparent and trustworthy. This course is a comprehensive, hands-on guide to Interpretable Machine Learning, empowering you to develop AI solutions that are aligned with responsible AI principles. You will also gain an understanding of the emerging field of Mechanistic Interpretability and its use in understanding large language models.

Through discussions, case studies, programming labs, and real-world examples, you will gain the following skills:

1. Describe interpretable machine learning and differentiate between interpretability and explainability.
2. Explain and implement regression models in Python.
3. Demonstrate knowledge of generalized models in Python.
4. Explain and implement decision trees in Python.
5. Demonstrate knowledge of decision rules in Python.
6. Define and explain neural network interpretable model approaches, including prototype-based networks, monotonic networks, and Kolmogorov-Arnold networks.
7. Explain foundational Mechanistic Interpretability concepts, including features and circuits.
8. Describe the Superposition Hypothesis.
9. Define Representation Learning and be able to analyze current research on scaling Representation Learning to LLMs.

This course is ideal for data scientists or machine learning engineers who have a firm grasp of machine learning but have had little exposure to interpretability concepts. By mastering Interpretable Machine Learning approaches, you'll be equipped to create AI solutions that are not only powerful but also ethical and trustworthy, solving critical challenges in domains like healthcare, finance, and criminal justice.

To succeed in this course, you should have an intermediate understanding of machine learning concepts like supervised learning and neural networks.

Coursera

If you have any Coursera-Specific questions about navigation or assignment types, the best resource would be the Coursera Learner Help Center. You can click the small question mark icon in the bottom right of your screen at any time to access help.

Certificate vs. Audit Experience

The required assignments for this course to earn the certificate are graded quiz activities.

You will only be able to submit graded quizzes if you are enrolled to earn the certificate, rather than auditing the course.

As a learner enrolled for a certificate, you will be able to access:

- The course quizzes, which means you will be able to submit the quizzes and receive automatic feedback on your answers.
- All other course content (videos, readings, labs, and discussions).

As an auditing learner, you will only be able to access the videos, readings, labs, and discussions. You can view the graded quizzes, but you will not be able to submit your answers and receive feedback.

[Go to next item](#)

Completed

Like

Dislike

Report an issue

Close X

Keep going!

Great job continuing your learning journey today. Every day of learning brings you closer to your goals.

